

Epic: Model Selection & Architecture

Project: LegalEase

1. Epic Overview

Select the AI model(s) and define the end-to-end architecture for LegalEase, an AI legal assistant, so that the product is accurate, secure, scalable and cost-effective.

2. Background & Problem Statement

In the legal domain, hallucination, data privacy and citation accuracy are critical. A wrong model or architecture choice can lead to incorrect legal output or leakage of confidential client data.

3. Goals & Objectives

- Shortlist and benchmark candidate LLMs on legal tasks
- Decide between RAG, fine-tuning or a hybrid approach
- Design a scalable, secure system architecture
- Define quality, latency and cost targets

4. Stakeholders

Role	Interest
Product Owner	Scope, priorities, fallback and human-review flow
Architect / Tech Lead	Model choice, architecture decisions
ML Engineers	Evaluation, RAG pipeline
Legal SMEs	Test data, answer quality, citations
Security / Compliance	Data privacy, residency, audit

5. Scope

5.1 In Scope

- LLM evaluation and selection
- RAG vs fine-tuning decision
- System architecture and data pipeline design
- Security and compliance design

5.2 Out of Scope

- UI/UX design
- Full production rollout
- Training a model from scratch

6. Assumptions & Constraints

- Legal SMEs will be available to help build the evaluation test set
- Client data must not be used by any provider for training
- Final choice must fit the approved infrastructure budget

7. Model Evaluation Criteria

Criterion	What we check
Accuracy	Correctness on legal Q&A, clause extraction, summarization
Hallucination rate	Unsupported or fabricated statements
Citation quality	Answers traceable to source documents
Latency	p95 response time
Cost	Cost per query at expected volume
Privacy	Data handling, retention, deployment options

8. Architecture Components

- **Ingestion:** document upload, parsing, chunking
- **Retrieval:** embeddings and vector database
- **Model layer:** primary LLM, fallback model, prompt management
- **Guardrails:** citation checks, confidence scoring, human review
- **Security:** encryption, access control, audit logging
- **Monitoring:** quality, latency and cost dashboards

9. User Stories

- 1 As an architect, I want to compare 3-4 candidate models on accuracy, latency and cost so that I can select the best fit.
- 2 As an ML engineer, I want to design a RAG pipeline (chunking, embeddings, vector DB) for legal documents.
- 3 As a lawyer, I want every answer to include source citations so that I can verify it.
- 4 As a security lead, I want client data encrypted and never used by the model provider for training.
- 5 As a product owner, I want a fallback and human-review flow for low-confidence answers.

10. Acceptance Criteria

- Benchmark report covering at least 3 models on a legal test set
- Targets defined for hallucination rate and citation accuracy
- Architecture diagram and decision record (ADR) approved
- Privacy/compliance checklist (data residency, retention) completed

11. Deliverables & Milestones

Phase	Deliverable
1. Discovery	Requirements, evaluation criteria, legal test set
2. Evaluation	Model benchmark report
3. Design	Architecture diagram and ADR
4. Review	Security/compliance sign-off
5. Handoff	Approved architecture for implementation epics

12. Risks & Mitigation

Risk	Mitigation
Hallucinated legal content	RAG with mandatory citations, human review
Vendor lock-in / cost spikes	Model-agnostic design, cost monitoring
Confidentiality / regulatory issues	Early security review, data residency controls

13. Dependencies

- Legal test dataset
- Security review
- Infrastructure budget approval

14. Success Metrics

- Answer accuracy
- Citation correctness
- p95 latency
- Cost per query

15. Open Questions

- Which jurisdictions and document types come first?
- Cloud-hosted API or self-hosted model?
- What accuracy and hallucination thresholds are acceptable?